Mono Ruzi

BACKEND ENGINEER · CRYPTO TRADING SYSTEMS, INTEGRATIONS, AND SEARCH

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SUMMARY

Backend engineer who builds the integration layer where money moves. Owns the exchange-facing execution path on a crypto platform handling $2M a month across eight exchanges, and ships TypeScript services on MongoDB and OpenSearch. Eight years across Python and Node.js, including the Kubernetes and AWS operations needed to run what is built.

EXPERIENCE

Finestel — **Senior Backend Engineer**

Jun 2024 – Present

Remote · Crypto copy-trading and trading automation · 40,000+ connected accounts · team of 6

- Owns the exchange-integration layer: one order-management dispatcher in front of eight exchange SDKs (Binance, Bybit, Bitget, OKX, KuCoin, Kraken, Gate, Hyperliquid), which keeps rate limiting, error normalization, and market-type guards in one place and each exchange's quirks in its own client.
- Built and owns the signal-bot execution engine, processing **~25K** TradingView webhook signals per day from order sizing and precondition checks through placement and post-fill callbacks.
- Concurrent fills were producing duplicate and mis-sized orders; Redis distributed locks around trade callbacks and balance updates brought order-duplication incidents to near zero. Self-healing reconciliation between exchange and local state cut stuck-position incidents needing manual intervention by **~75%**.
- Eliminated exchange API bans with a Redis sorted-set sliding-window rate limiter, throttling per account and per action type below each exchange's ceiling.
- Delivered the commercial integrations on the main API: Stripe subscription billing, a white-label partner API serving 15 partners, HubSpot CRM sync, and Hyperliquid agent-wallet and builder-fee support.
- Cut p95 on the heaviest list endpoints from **2.1s to 380ms** through select_related and prefetch_related restructuring and an in-process TTL cache for hot reference data.
- Schema changes on the 60M-row MariaDB trade table are forced through ALGORITHM=INSTANT, LOCK=NONE, so an ineligible change fails fast instead of locking writes.
- Wrote the architecture documentation and about 20 domain rule documents the team works from; GitLab CI gates on ruff and Django system checks.

Independent — **Technical Consultant**

2019 – Present

Remote · Freelance, concurrent with the roles below

- Advises 10+ companies on architecture and delivery, with AWS infrastructure (EC2, S3, RDS, Lambda) defined in Terraform, CloudFormation, and Ansible.

Dirichlet — **AI Platform Engineer**

May 2026 – Sep 2026

Stockholm, Sweden (Remote) · Part-time · Traffic-arrangement planning platform

- Led the design of an LLM autofill pipeline that turns a work-order email plus geospatial context into a Swedish regulatory survey and a 22-item risk matrix, reducing a three-hour manual step to **under 15 minutes** for 40+ planners.
- Defined a strict JSON schema over the OpenAI Responses API with a deterministic post-processor, so model output cannot leave the approved option set. SHA-256-keyed Redis caching removed **~60%** of repeat calls to upstream geodata APIs.
- Built RBAC end to end on Supabase phone-OTP JWT authentication: roles, permissions, database triggers, an immutable super_admin role, and Redis-backed admin impersonation for support debugging.

Yarai — **AI Platform Engineer**

Dec 2025 – May 2026

Tehran · Contract, part-time · Self-hosted LLM platform

- Built a self-hosted inference platform on Kubernetes, KServe, and vLLM serving five production models up to 27B parameters behind a LiteLLM gateway, delivered through GitLab CI, Harbor, and ArgoCD on bare-metal clusters.

Hamravesh — **Site Reliability Engineer** Sep 2023 – Jun 2024

Tehran · PaaS platform

- Provisioned multi-tenant S3 on SeaweedFS across a **100+ node** cluster holding **40+ TB**, with a FastAPI provisioning API and asynchronous replication to MinIO and AWS S3.

Mofid Securities — **Backend Developer** Jan 2022 – Sep 2023

Tehran · Bourseview, Iran's largest stock market data platform (~1 TB)

- Built the BV-Chart microservice (FastAPI, TradingView standard) on ClickHouse and cut historical query time by **50%**.
- Diagnosed Gunicorn workers exiting on response timeout. A Sentry decorator on the JRPC endpoints exposed an N+1; removing it made that endpoint **10×** faster.

Homeca.ir — **Tech Lead** 2019 – 2022

Tehran · At-home health services marketplace

- Owned the marketplace architecture as technical lead: a Django REST Framework backend with row-level locking for transactional integrity, Celery and RabbitMQ for async work, and SMS OTP authentication.
- Shipped an SEO-optimized Next.js static CMS behind Cloudflare (A+ GTmetrix). GitLab CI gated rolling Kubernetes updates on unit tests, holding production deploys at zero downtime.

Mizan Gostaran Sharif — **Web Developer** Feb 2018 – Jul 2019

Tehran · HR evaluation and ranking platform

- Built Express.js services on PostgreSQL, with MongoDB as the log store, and the React and Next.js frontend consuming them; Swagger/OpenAPI specifications kept the REST contracts consistent across teams.

PROJECTS

Datasource Management Panels Aug 2026 – Sep 2026

TypeScript, Next.js, Node.js, MongoDB, OpenSearch, PostgreSQL, MinIO · four consoles in production on Vercel

- The OpenSearch console flattens nested mappings into dotted field paths and compiles each filter into the query its field type needs: simple_query_string across text fields, term on the .keyword subfield, match_phrase, wildcard, and range.
- The MongoDB console infers a typed schema from a document sample and compiles the query UI to native Mongo, using $text where a text index exists and an escaped-regex fan-out where it does not, with server-side pagination throughout.

Optimalist Mar 2024 – Present

TypeScript, Bun, MongoDB, OpenSearch, Pinecone · solo, in production

- Built an agentic content loop: actions in a Telegram quiz launch cloud coding agents against the course repository, and each merged pull request is synced and served with no manual authoring step.
- A Sentry-compatible ingest endpoint indexes error events into daily OpenSearch indices; **220** tests cover the service.

SKILLS

Languages	TypeScript, JavaScript, Python, SQL, Bash, Go (maintenance)
Backend	Node.js, Express.js, Bun, Django, Django REST Framework, FastAPI, Celery, Pydantic, REST API design
Data and search	MongoDB, MySQL / MariaDB, PostgreSQL, Redis, ClickHouse, OpenSearch (query DSL, mappings), Pinecone
Integrations	Exchange APIs (Binance, Bybit, OKX, Hyperliquid), Stripe, HubSpot, TradingView webhooks, GitHub REST API
Infrastructure	AWS (EC2, S3, RDS, Lambda), Kubernetes, Docker, Terraform, ArgoCD, Cloudflare, Sentry, Prometheus
AI	Cloud coding agents, OpenAI API structured output, LLM grounding, vLLM, LiteLLM
Spoken	Persian (native), English (advanced), Turkish (intermediate)

EDUCATION

Sharif University of Technology — **M.Sc. Bioelectrical Engineering** 2018 – 2020

Sharif University of Technology — **B.Sc. Electrical Engineering** 2014 – 2018